Supplementary Material

Collusion-resilient, tamper-evident range analytics over homomorphically encrypted IoT data in multi-cloud storage

Section, theorem, table and equation numbers without an "S" refer to the main text. Tables and figures here are numbered S1, S2, ...

S1. Why a homomorphic layer at $t = n - 1$: the full argument

This section completes Section 4.1 of the main text.

The cross terms are not an artefact of choosing linear answers. Take any non-interactive scheme in which CP_1 holds an additive share S_1 of the data, the client forms its queries without reference to the data, providers answer independently, and every query and response is shorter than a share. Given a database X held by a single server, a client can run that scheme with $S_1 := X$ and the other shares drawn by itself, which fixes the data as $A = X + \sum_{k>1} S_k$. It plays providers $2, \ldots, n$ locally, sends only the first query to the server, and subtracts $\sum_{k>1} S_k[\alpha]$ from the result. Privacy against the single provider CP_1 hides α from the server, so this is a non-trivial single-server PIR scheme for X, and such schemes imply oblivious transfer [1]; their standard instantiation is homomorphic encryption [2]. Within that class, secret-sharing the data does not remove the public-key primitive, it moves it into a per-share PIR; the alternatives are interaction, for example multiplication triples, which the no-channel constraint excludes, or assumptions outside the class, such as preprocessing. The reduction does not show that every secret-sharing system needs homomorphic encryption. The argument needs the scheme to be correct on every tuple of shares, which holds for any scheme whose answers are a deterministic function of the shares and the queries, and it applies at $t = n - 1$ with (n, n) sharing. A lower threshold escapes it: with $(t{+}1, n)$ threshold shares and $t < n/2$, providers can multiply two shared values locally and the client interpolates the degree-$2t$ result, so low-degree selection needs neither interaction nor a public-key primitive. That is the route of Obscure [3], whose share multiplications raise the polynomial degree, so that one-round queries need $2l + 1$ servers for a predicate of length l and the adversary must stay below a majority [3, §2.1, §3.2]. A store replicated under one owner's homomorphic key is the direct way to supply the primitive at $t = n - 1$, and it also brings three operational properties that a shared store lacks.

Durability. A sharing with privacy threshold $n-1$ is lost when any one share is lost, whereas a replica set survives the loss of $n - 1$ providers, which is the reason multi-cloud deployments replicate at all. Threshold sharing does not escape the trade-off: a $(t{+}1, n)$ scheme tolerates $n - t - 1$ losses, so at $t = n - 1$ it degenerates to the (n, n) case.

Graceful degradation. A query needs every provider in the set it is addressed to. With replicas, queries continue over any surviving set of at least two providers, at that set's collusion bound, without touching stored state; a shared store cannot answer until the dealer re-shares.

Key management and churn. Encryption under one owner key lets the owner rotate keys, revoke a provider, or add one by copying ciphertexts. Under sharing, every change to the provider set re-deals the whole store, and the dealer must hold the plaintext to do it.

Remark S1 (Stripping the mask first)**.** *An alternative contracts the masked shares b_j directly against the record column and cancels the resulting term $u_f(\alpha_w) = \sum_i w_f[i]\, P[\alpha_w][i]\, m_i$ with an encrypted correction column $U_f = (\mathsf{Enc}_{pk}(u_f(x)))_{x\in[N]}$, retrieved by the same DPF. An earlier version of our design did this. It has two defects. U_f must be precomputed by the DO for each weight vector, so it serves only registered weights, for which the fast path is cheaper; it cannot serve the ad-hoc weights and conjunctions it was meant for. And the correction must itself be encrypted: published in the clear, the count correction $g(x) = \sum_i P[x][i]$ gives $\sum_i I'[x][i] - g(x) = |I[x]|$, the row Hamming weight, restoring the volume leakage the mask removes. The flexible path avoids both by stripping the mask at the client before any contraction.*

S2. Tamper evidence: verification modes and full proof

One owner key. Row 1 holds δm_i, or $\delta A_f[x]$, under a single secret δ, and the DU accepts only if $\hat{y} = \delta y$. A provider that adds a constant, corrupts one half, or returns a stale or foreign ciphertext is caught except with probability $1/p$. One that applies the *same* linear substitution to both halves passes: contracting against any e'_j instead of e_j, for instance $k\, e_j$, shifts y and $\hat{y}$ by amounts in the ratio δ, so the tag still verifies.

Per-position keys. Row 1 holds $\delta_x A_f[x]$ with $\delta_x = F''_{K''}(x)$, or $\delta_c T_f[c]$ per cube cell, and the DU checks against the key of the queried position. A substitution $e_j + d$ then passes only if $\sum_{x \neq \alpha_w} d[x](\delta_x - \delta_{\alpha_w}) A_f[x] = 0$, a non-zero linear form in independent keys whenever d touches a non-empty position other than α_w. What survives is a deviation concentrated on the queried position itself, which rescales the answer; a provider that guesses or knows that position wins. Per-position keys also do not exist for the flexible path, whose contraction runs over records.

A per-query secret. The session layout stores no key at all: row 1 of every column, aggregate column or table is a copy of row 0. With each query the DU draws a fresh δ_s uniformly from $\mathbb{Z}_p^*$ and sends a second set of keys for the same points with outputs multiplied by δ_s: f_{α_w, δ_s} on the fast path, $\delta_s \beta_\sigma$ at each corner of a range, and fresh shares of $\delta_s v$ in a flexible round. Each provider places the second evaluation vector in row 1 of the same plaintexts, so the homomorphic work is unchanged and slot $\ell/2$ decrypts to $\delta_s y$. The DU accepts only if $\hat{y} = \delta_s y$.

Theorem 2 (Tamper evidence). *Consider one query in the session layout on any path. Let a coalition C of at most $n-1$ providers return arbitrary responses computed from its members' views, with at least one honest provider, an honest DU that keeps δ_s secret, and no provider colluding with the DU; the coalition does not see the honest providers' keys, and the DU decrypts whatever ciphertexts it receives. Let y be the honest answer. The DU accepts an output $y' \neq y$ with probability at most*

$$\frac{1}{p-1} + \varepsilon,$$

where $\varepsilon = \kappa \operatorname{Adv}^{\mathrm{dpf}}$ for a query whose δ_s-carrying set has κ DPF keys per provider ($\kappa = 1$ on the fast path and for a retrieval, 2^d for a range) and $\varepsilon = \operatorname{Adv}^{\mathrm{prf}}$ for a flexible round. Over Q queries the bounds add, whether or not the coalition learns which queries were accepted.

Proof. Let V be the coalition's view: both key sets of its members, the stored ciphertexts, the public and evaluation keys, and everything it saw in earlier queries. Its responses decrypt, summed over C, to (r_0, r_1) in the slots of one template, a function of V, the database and the coalition's coins; the decryption key is independent of δ_s. Let y_C and $\hat{y}_C$ be what the members of C would have contributed honestly, also functions of V and the database. The honest providers contribute $y - y_C$ and $\delta_s y - \hat{y}_C$, so the DU decrypts $y' = y - y_C + r_0$ and $\hat{y}' = \delta_s y - \hat{y}_C + r_1$ and accepts if and only if

$$r_1 - \hat{y}_C = \delta_s\,(y' - y).$$

Replace the coalition's δ_s-carrying keys by keys for the same points and an independent scalar δ'_s. A reduction that knows the points, the signs, the database and δ_s can evaluate the event $\{y' \neq y \wedge r_1 - \hat{y}_C = \delta_s(y' - y)\}$ from the coalition's keys and outputs, so by $(n-1)$-privacy, which hides the output β as well as the point, its probability changes by at most $\kappa \operatorname{Adv}^{\mathrm{dpf}}$; for a flexible round the coalition's shares of $\delta_s v$ are uniform, or pseudorandom when a seed is missing, at cost $\operatorname{Adv}^{\mathrm{prf}}$. After the change V is independent of δ_s. Conditioned on V, the database and the coins, everything in the displayed equation except δ_s is fixed, and since $y' - y \neq 0$ at most one of the $p-1$ equally likely values of δ_s satisfies it. Each query uses a fresh δ_s, so feedback from earlier queries concerns independent scalars, and a union bound over the Q queries completes the proof. □

The bound needs no restriction on how a provider deviates: it may substitute vectors, rescale the queried point, swap or mix slot rows, replay an old response or return any ciphertext at all. It does not guarantee output delivery, since a deviating provider can always force an abort. It also leaves one channel open: whether the DU aborts can depend on the data, for example on whether a cell a provider chose to perturb is empty, so a DU that reveals its aborts can leak up to one bit about the data per query to a malicious provider, though nothing about the query, whose keys are pseudorandom to it. A DU that retries silently through another provider set closes that channel. The 200 injected trials per deviation in Section 7.3 are empirical checks of the implementation, not a measurement of the $1/(p-1)$ bound. Against per-position keys the cost is a second key set and a second full-domain evaluation per provider, with no extra homomorphic product and no stored key; Section 7.3 measures all three modes on every path.

S3. Correctness: full proof

Theorem 3 (Correctness). *Fix a database DB, a keyword w, and a registered template f satisfying the magnitude bound of Lemma 1. Then the data user's combination step returns $y = \sum_{i \in S} w_f[i]\, m_i$ with $S = \{i : I[\alpha_w][i] = 1\}$ on the fast path, a range query on a box B of a registered cube whose total is below p returns $\sum_{i:\, c(i) \in B} w_f[i]\, m_i$, a retrieval returns exactly S, and a flexible round on a selection vector v satisfying the magnitude bound returns $\sum_i v[i]\, m_i$, except with the negligible probability that leveled-HE decryption fails.*

Proof. By DPF correctness $\sum_j e_j[x] = 1$ at $x = \alpha_w$ and 0 elsewhere.

Fast path. $\boxplus_j \mathrm{ct}_j$ decrypts, by Lemma 1 applied with $\beta = \left(\sum_j e_j\right)$ over the array A_f, to $A_f[\alpha_w] = \sum_i I[\alpha_w][i] w_f[i] m_i = \sum_{i \in S} w_f[i] m_i$.

Range. The 2^d keys of a range query share the point functions $\beta_\sigma\, u_{\kappa_\sigma}$, with $\beta_\sigma = 0$ for a corner outside the cube. In the combined layout $\sum_j e_j = \sum_\sigma \beta_\sigma u_{\kappa_\sigma}$, so by Lemma 1 applied over the column T_f the response decrypts to $\sum_\sigma \beta_\sigma T_f[\kappa_\sigma]$, and in the verified layout row 1 decrypts to δ_s times the same sum; in the per-corner baseline segment σ decrypts to $\beta_\sigma T_f[\kappa_\sigma]$ and the DU adds them. By (3) that sum is the box aggregate modulo p. The true aggregate lies in $[0, p)$ because it is at most the cube's total, so the reduction loses nothing, even though individual signed terms are negative.

Retrieval. $\sum_j b_j = \sum_x \left(\sum_j e_j[x]\right) I'[x] = I'[\alpha_w] = \mathrm{pack}(b) + P[\alpha_w]$, so $\left(\sum_j b_j\right) - P[\alpha_w] = \mathrm{pack}(b)$ exactly, and the DU, which derives $P[\alpha_w]$ from K', unpacks b.

Flexible round. $\sum_j v_j = v$, so by Lemma 1 applied with $\beta = v$ over the record column, $\boxplus_j \mathrm{ct}_j$ decrypts to $\sum_i v[i]\, m_i$ in slot 0 and $\delta \sum_i v[i]\, m_i$ in slot $\ell/2$.

Packing. Rotations act within slot rows and the rotate-and-sum covers one segment, so the slot that carries template r receives only the products of template r's segment, and the value and MAC rows do not mix.

The dictionary is injective by construction, so row α_w is exactly the records containing w. Decryption correctness under the noise growth discussed in the proof of Lemma 1 contributes the negligible failure term. The check of Section 4.6 passes with certainty for honest providers: in the session layout row 1 holds the same data and the second key set shares δ_s times the same point functions or vector, so by the same argument slot $\ell/2$ decrypts to $\delta_s y$; with stored keys, δ or δ_{α_w} distributes over the linear contraction, $\sum_x e[x]\, \delta_x A_f[x] = \delta_{\alpha_w} A_f[\alpha_w]$ when $e = u_{\alpha_w}$. □

S4. Leakage security: definition and full proof

Leakage profile. We define $\mathcal{L} = (\mathcal{L}_{\text{Setup}}, \mathcal{L}^{\text{CP}}, \mathcal{L}^{\text{DU}})$ with an observer split:

$$\begin{aligned}\mathcal{L}_{\text{Setup}}(DB) &= (n, t, N, N_d, p, \ell, \text{layoutMeta}, \text{cubeMeta}, \text{roleMeta}, \text{tierMeta}),\\ \mathcal{L}^{\text{CP}}_{\text{Query}} &= \text{class}(q),\\ \mathcal{L}^{\text{DU}}_{\text{Query}} &= \big(|S|,\ f(\text{records}(S))\big).\end{aligned}$$

The setup leakage is everything a provider can see in its replica: layoutMeta gives the registered templates and the segment and ciphertext-column shapes, cubeMeta the registered cubes and their dimensions $(s_1, \ldots, s_d)$, roleMeta the number of roles and each role's span of packed words, and tierMeta the tier sizes and epochs. Here $\text{class}(q)$ says whether a query is a fast-path aggregate over a named layout of registered templates, a range aggregate over a named cube and layout, a retrieval, or the second round of a flexible query, together with the querying user's role and whether the query carries the verification key set. The coalition learns that and the public dimensions of the deployment, and nothing else: not the keyword, not the range or whether it touches the cube's edge, not the access pattern, not $|S|$, not the weights, not the result. In contrast to a design in which the matched set is returned to the providers for evaluation, and in contrast to an unmasked index, $\mathcal{L}_{\text{Setup}}$ here contains no posting-list lengths and no membership structure, by Theorem 1. Retrieval adds the public padding bound $k_{\max}$ to $\mathcal{L}^{\text{CP}}$. A conjunctive query is two retrievals and one flexible round, adding only result sizes to $\mathcal{L}^{\text{DU}}$, never to the coalition.

Definition 1 ($\mathcal{L}$-security against t-collusion). *In* $\mathbf{Real}_{\mathcal{A},\Pi}(1^\lambda)$*: (1)* $\mathcal{A}$ *outputs* (DB, C) *with* $|C| \le t$*; (2) the challenger runs the outsourcing of Algorithm 1 and gives* $\mathcal{A}$ *the static view* $(I', CT, \widehat{CT}, \{A_f, \widehat{A}_f\}_f, \{T_f, \widehat{T}_f\}_{f,cube}, pk, pp)$*; (3) for* $q = 1, \ldots, Q$*,* $\mathcal{A}$ *adaptively issues a query (a keyword with a template layout, a box of a registered cube with a layout, a keyword to retrieve, or a selection vector* v_q *for a flexible round) and receives the coalition's per-query view: its DPF keys* $\{\kappa_{q,j}\}_{j\in C}$ *or its shares* $\{v_{q,j}\}_{j\in C}$*, including the* δ_s*-carrying set of a verified query, and the responses those providers compute; (4)* $\mathcal{A}$ *outputs a bit. In* $\mathbf{Ideal}_{\mathcal{A},\mathcal{S},\mathcal{L}}(1^\lambda)$ *a* PPT *simulator* $\mathcal{S}$ *produces the views given only* $\mathcal{L}_{\text{Setup}}$ *and the per-query leakage.* Π *is* $\mathcal{L}$*-adaptively-secure against* t*-collusion if for every* PPT $\mathcal{A}$ *corrupting* $\le t$ *providers there is a* PPT $\mathcal{S}$ *with* $|\Pr[\mathbf{Real} = 1] - \Pr[\mathbf{Ideal} = 1]| \le \mathsf{negl}(\lambda)$.

Theorem 4 (t-collusion leakage security). *If* F *and* F' *are secure PRFs,* $(\mathsf{Gen}, \mathsf{Eval})$ *is an* $(n-1)$*-private DPF, and the homomorphic and symmetric schemes are IND-CPA, then CR-SHE is* $\mathcal{L}$*-adaptively-secure against* t*-collusion in the sense of Definition 1, for* $t = n - 1$ *and* $\mathcal{L}^{CP}_{\text{Query}} = \text{class}(q)$*. Concretely, for every* PPT $\mathcal{A}$ *there exist* PPT $\mathcal{S}$ *and* $B_1, \ldots, B_4$ *with*

$$\begin{aligned}\big|\Pr[\mathbf{Real}{=}1] - \Pr[\mathbf{Ideal}{=}1]\big| \le\ & \text{Adv}^{\text{prf}}_{F',B_1}\\ &+\ Q \cdot \big(2^{d_{\max}+1}\text{Adv}^{\text{dpf}}_{B_2} + 2\,\text{Adv}^{\text{prf}}_{B_4}\big) +\ m \cdot \text{Adv}^{\text{cpa}}_{B_3},\end{aligned}$$

where m *is the number of ciphertexts in the coalition view. As is standard for BFV, IND-CPA is assumed to hold even given the relinearisation and rotation keys the providers need (circular security). The theorem is stated for one static tier; with a fresh* K' *per tier (Section S5.1) it applies to each tier independently, with one PRF term per tier.*

Proof. Take $|C| = n-1$ without loss of generality; a smaller coalition holds a subset of the same state. Let $H_0 = \mathbf{Real}$.

H_1 (PRF to random, index to zero). Replace F' by a truly random function, at cost $\text{Adv}^{\text{prf}}_{F',B_1}$. The mask is then an exactly uniform one-time pad (Theorem 1), so building I' from the all-zero index instead of I changes nothing; I' now depends only on (N, N_d).

H_2 *(DPF keys to simulated).* Replace every DPF key the coalition holds by $\mathcal{S}_{\mathrm{DPF}}(1^\lambda, C, \cdot)$ for its domain, licensed by $(n-1)$-privacy, which hides the output β as well as the point. A range query to a d-dimensional cube carries 2^d independent keys, dummy keys included, and 2^d more when it is verified, so a hybrid over all keys costs $Q\, 2^{d_{\max}+1}\mathrm{Adv}^{\mathrm{dpf}}_{B_2}$, where $d_{\max}$ is the largest registered cube dimension. In particular the simulated keys no longer depend on δ_s. The per-query key material now depends only on C, the domain and the query class; in particular it no longer depends on which corners were dummies.

H_2' *(flexible shares to uniform).* In each flexible round at least one share lies outside C. If it is the explicit difference vector, the coalition's shares are seed expansions independent of v_q. If it is a seed's expansion, the explicit vector is v_q (and $\delta_s v_q$ in the verified layout) minus a PRF output the coalition cannot compute, and replacing both outputs by uniform vectors costs $2\,\mathrm{Adv}^{\mathrm{prf}}_{B_4}$. Either way the coalition's shares become uniform and independent of (v_q, δ_s), over at most Q rounds.

H_3 *(ciphertexts to encryptions of* 0*).* Replace each of the m ciphertexts in the coalition view, namely the record column, the columns $A_f, \widehat{A}_f$ and the tables $T_f, \widehat{T}_f$, by encryptions of 0 under pk, at cost $m \cdot \mathrm{Adv}^{\mathrm{cpa}}_{B_3}$, since the coalition holds only pk.

Simulator. The responses $\mathrm{ct}_{q,j}$ need not be simulated separately: each is a deterministic function of $\kappa_{q,j}$ or $v_{q,j}$, the query class, and the provider's stored state, all of which $\mathcal{S}$ possesses in H_3. Compression is a public, deterministic map and changes nothing. $\mathcal{S}$ therefore regenerates the entire coalition view from $\mathcal{L}_{\mathrm{Setup}}$ and class(q) alone, so $\mathbf{Ideal}_{\mathcal{A},\mathcal{S},\mathcal{L}}$ matches H_3; summing the terms proves the theorem. □

Remark S2 (Rounds and leakage)**.** *Suppose the client recovers S and then requests* $\mathsf{Compute}(pk, \{CT_i\}_{i\in S}, f)$ *from the providers. The coalition's per-query view then contains S itself, which is a function of the queried keyword and the database. No simulator given only data-independent leakage can produce it, so an empty per-query leakage profile is* unattainable *in that design rather than merely loose. The flexible path has the same two rounds, but its second round sends fresh shares of the selection instead of the selection, and H_2' shows those are simulatable. What makes Theorem 4 true is contracting a share at the provider, not the number of rounds.*

Remark S3 (Confidentiality and degradation)**.** *For two databases of equal public dimensions differing only in contents, Theorem 4 makes each coalition view indistinguishable from a distribution depending only on $\mathcal{L}_{\mathrm{Setup}}$, which is identical for both; the executions differ by at most $m \cdot \mathrm{Adv}^{\mathrm{cpa}}$, with the content dependence isolated in H_3. The same proof applies for any $|C| \le n-1$, since $(n-1)$-privacy holds for every strict subset; a coalition of all n providers is outside the model, and against it the guarantee degrades to IND-CPA confidentiality of the records with the index still masked under F'.*

Remark S4 (Beyond semi-honest)**.** *Theorem 4 is proved for semi-honest providers. Because providers never interact and receive nothing but keys or shares, a coalition that deviates sees the same view, so its deviation buys it no information about the query, and Theorem 2 makes any deviation that changes an answer evident to the DU. Two things are not covered: output delivery, since a deviating provider can force an abort, and the abort events themselves, which can carry up to one bit about the data per query if the DU reveals them (Section 4.6). This is why Table 1 records* abort *rather than full malicious security.*

S5. Additional design details and results

S5.1 Dynamic ingest

IoT streams are append-only. CR-SHE keeps a static tier and a hot buffer of at most B records scanned by the same contraction, folded into a rebuilt tier every B insertions. Because the coalition's per-query view is independent of the query (Theorem 4), a query reveals nothing

about records inserted before or after it, which is forward and backward privacy in the usual sense; what updates themselves expose is the public tier metadata in the setup leakage, and we do not analyse update leakage beyond that. Three rules keep updates from leaking. *Rows never move*: a new keyword takes the next free, masked, all-zero row of D, so only exhausting the provisioned headroom is visible. *A fresh mask key per static tier*: a rebuild re-lays the record axis, and reusing K' would expose the change in membership. *Whole columns are replaced*: re-encrypting only the affected ciphertexts of A_f would reveal which blocks of tags changed; summed-area tables with time as the outermost axis would be the exception (Section 4.5), since only the newest, publicly known slice changes. The C++ prototype implements and measures the hot buffer and the fast-path rebuild (Section S5.11); the reference implementation does not, and neither implements the summed-area-table update.

S5.2 Complexity analysis

Table S1 summarises the asymptotic costs. Four entries deserve comment because they correct the intuitive accounting.

First, the selection contraction of a retrieval, and hence of the flexible path's first round, is $\Theta(NN_d/\pi)$ ring operations ($\pi = 32$ packed bits per word, $\Theta(NN_d)$ unpacked), not $\Theta(N)$: a full-domain evaluation produces N share values, each of which multiplies an index *row* of $\lceil N_d/\pi \rceil$ packed words. Counting only the $O(N)$ pseudorandom-generator calls of the evaluation omits the dominant term. With a small modulus and vectorised arithmetic these are cheap word operations, and the pass is embarrassingly parallel, but the asymptotic is what it is: this is the ORAM and PIR lower-bound price for hiding the access pattern from a coalition without preprocessing [4, 5]; preprocessing PIR avoids it at the cost of client hints that updates invalidate [6].

Second, the fast path of Section 4.4 removes the N_d factor entirely, at $O(N)$ generator calls plus $O(N/\ell)$ packed ciphertext operations, at the cost of restricting queries to single keywords and templates registered at setup.

Third, the aggregate response is one ciphertext, for up to $\ell/2$ templates, and, crucially, *independent of* $|S|$. A response whose size grew with the result set would leak the result-set size to the coalition and contradict $\mathcal{L}^{\mathrm{CP}}$. Retrieval responses are $\lceil N_d/32 \rceil$ words of fixed size for the same reason. Range queries likewise return one ciphertext, for any box.

Fourth, the single-server alternative of Section 4.4 needs no non-collusion assumption but pays in two places. On the fast path its query is $\lceil N/(\ell/2) \rceil$ fresh ciphertexts instead of an $O(\lambda \log N)$ key, and its contraction uses ciphertext-ciphertext products. On retrieval and the flexible path it must multiply an encrypted N-vector by the packed $N \times \lceil N_d/32 \rceil$ index, about $N\lceil N_d/32 \rceil/\ell$ plaintext-ciphertext products with diagonal packing [7], where CR-SHE does $\Theta(NN_d/32)$ word operations. Both are linear in NN_d: the gap is the constant ratio of a homomorphic slot operation to a machine-word operation, which Section S5.6 bounds from below. What does not shrink with better kernels is the client side: a $\lambda \log N$-bit key against $\lceil N/(\ell/2) \rceil$ fresh encryptions.

S5.3 Range layouts and cube size on a synthetic cube

The measurements below use a synthetic cube with the telemetry dataset's dimensions, $3 \times 11{,}523 \times 10$ ($M = 345{,}690$ cells), on the Intel Xeon VM, with the same queries as Section 7.2.

Cost does not depend on the range (Table S2). From a one-minute window to the whole 11,523-minute timeline, a provider spends 0.78–0.91 s in the combined layout, 1.40–1.53 s in the verified layout and 3.24–3.44 s in the per-corner layout. Every query receives one 132,081 B ciphertext. The whole-timeline box already starts at the cube's edge, so four of its eight corners are dummies; a 240-minute box at the cube's origin, on which seven of the eight are dummies,

Table S1: Asymptotic costs. γ is the per-provider DPF key size, $O(\lambda \log N)$ at $n = 2$; ℓ is the SIMD slot count.

Resource	Per provider	Data user
Index storage	$O(NN_d \log p/32)$, packed	n/a
SAT storage, d-cube	$O(kM/\ell)$ per role	n/a
Ciphertext storage	$O(N_d/\ell) + O(N/\ell)$ per f	n/a
Query uplink	$O(\gamma)$	$O(n\gamma)$
Flexible uplink	λ, or $N_d\lceil \log p \rceil$ bits	$O(N_d \log p)$
Aggregate downlink	1 ct	n ct
Retrieval downlink	$\lceil N_d/32 \rceil$ words, fixed	$O(nN_d)$ bits
Fast path	$O(N)$ PRG, $O(N/\ell)$ HE	1 decrypt
Range, d-cube, M cells	$2^{d+1} \cdot O(M)$ PRG, $O(kM/\ell)$ HE	2^{d+1} keygen, 1 decrypt
Range uplink (verified)	$2^{d+1}\gamma$	$n2^{d+1}\gamma$
Retrieval / round one	$O(N)$ PRG, $\Theta(NN_d/32)$ ring	$O(N_d)$
Flexible round two	$O(N_d/\ell)$ HE, $O(\log \ell)$ rot.	1 decrypt
Owner precomputation	postings per f; $O(NN_d)$ mask	n/a
Insertion, τ tags	$O(\tau)$ updates, $O(N/\ell)$ re-enc.	n/a

Table S2: Range aggregation per provider against window length L, $M = 345{,}690$ cells, $d = 3$, two templates. Reference implementation, one thread. Every query returns 132,081 B; the uplink per provider is 2,760 B (8 keys), or 5,520 B (16 keys) when verified.

L (min)	Per-corner	Combined	Verified (session)	
	total (ms)	total (ms)	DPF eval (ms)	total (ms)
1	3290.9	891.7	1055.0	1410.3
60	3436.1	913.7	1152.4	1437.5
240	3240.6	781.5	1143.9	1395.3
1,440	3253.3	810.2	1265.9	1527.2
11,523	3298.1	843.7	1103.4	1471.5

costs the same (937.8 ms combined, 1,385.0 ms verified) and sends the same number of keys. The data user spends 6–19 ms generating keys and 1.1–1.9 ms combining, checking and decrypting.

What verification costs. The verified layout stores the same 169 ciphertexts as the combined layout, with a copy of the table in place of a MAC row, and performs the same homomorphic products; what it adds is the second set of eight keys, 2,760 B more uplink, and their full-domain evaluation. Evaluating both key sets takes 1.06–1.27 s of its total, against 0.45–0.51 s for the combined layout's single set. It is 1.6–1.9 times the combined layout and less than half the per-corner layout, which stores the table eight times, contracts 1,351 chunks against 169, and still misses deviations at the queried corner (Section 7.3). At the measured 0.092 ms per plaintext-ciphertext product the products themselves account for 16 ms in the combined and verified layouts and 125 ms in the per-corner layout, a computed figure; the rest is DPF evaluation and the Python-side encoding of one plaintext vector per chunk, which a native implementation largely avoids.

Table S3: Range aggregation against cube size, $d = 2$, combined layout, one template. Reference implementation, one thread.

Cells M	Chunks	DPF eval (ms)	Provider total (ms)
10,000	3	8.3	34.7
99,856	25	60.9	129.3
1,000,000	245	722.2	1356.0

Cost grows with the cube, linearly (Table S3): 34.7, 129.3 and 1,356.0 ms per provider for 10^4, about 10^5 and 10^6 cells, so cube size, not record count or range length, is the quantity a deployment provisions. The owner chooses which families form a cube, as it chooses templates.

Against the alternatives. The same query through single-tag aggregates, with each cell registered as its own tag, needs one query per cell in the box: 1,200 queries for a 240-minute window over five bands, each with its own keys and response, and the number of queries reveals the box's size. Through the flexible path it needs a retrieval for the device, each of the 240 minutes and each band before the contraction. A single server holding the same encrypted table needs no non-collusion assumption but an encrypted query as long as the table: 169 fresh ciphertexts in the combined layout, 44.4 MB of uplink computed from the measured ciphertext size, against 2.8 kB of DPF keys per provider, or 5.5 kB verified.

S5.4 Selection latency and key size

Table S4 reports the per-provider full-domain evaluation in the C++ prototype, which is the whole of the symmetric-key selection. At $N = 11{,}580$ it costs 0.336 ms on one thread; at $N = 10^6$, 45.7 ms on one thread and 18.8 ms at thirty-two, beyond which it regresses as thread dispatch outweighs the per-level work. The two-party key grows from 266 B to 368 B, as $O(\lambda \log N)$ predicts. Sub-millisecond cells vary by up to 25% between runs, so only orders of magnitude are read from them: selection is not the bottleneck; the homomorphic contraction is.

Table S4: Per-provider selection latency and two-party key size against N. Medians of 21 runs.

N	Latency (ms), T threads					Key
	$T = 1$	4	16	32	64	(B)
1,000	0.032	0.032	0.032	0.032	0.032	198
10,000	0.290	0.175	0.156	0.161	0.183	266
11,580	0.336	0.200	0.177	0.181	0.211	266
100,000	4.418	2.487	1.909	1.928	2.420	317
10^6	45.741	25.701	19.711	18.769	24.401	368

S5.5 Oblivious aggregation

In the C++ prototype the fast path costs 38.8–39.2 ms per provider across a 405× range of record counts, $N_d = 10^3$ to 405,184 (Table S5, Fig. S1); an earlier sweep in increasing N_d showed a warm-up step near 53 ms that vanished when the order was reversed. The single-round general path of Remark S1, whose $\Theta(NN_d)$ pass is the flexible path's first round, cost 103.7 ms at $N_d = 10^3$ and 1,378.2 ms at the full corpus, 35.5× the fast path, with an unpacked masked index of 37.5 GB per provider, which packing divides by 32. The two-ciphertext response, 525,950 B, was byte-identical for an empty posting list and one matching 4,999 of 5,000 records; packed and compressed it is 132,081 B.

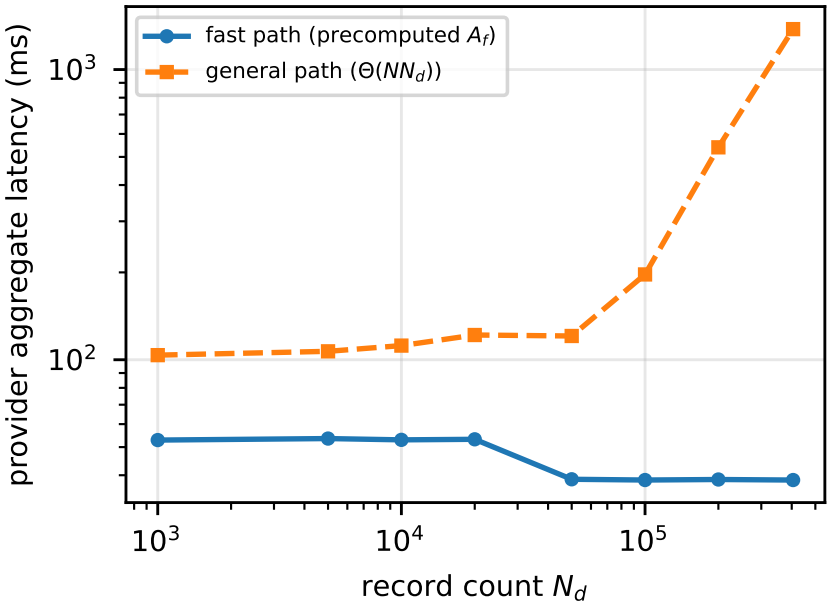


Figure S1: Aggregate latency, fast path against the single-round general path (C++, two-ciphertext layout). The x-axis is N_d, not $|S|$: cost no longer depends on the matched-set size. The fast path is flat to ±0.5% over a 405× range.

Table S5: Aggregate latency and storage, fast path against the single-round general path, $N = 11{,}580$, template SUM, C++ prototype, two-ciphertext layout. Medians of 11 runs (fast) and 3 (general).

N_d	Fast (ms)	General (ms)	I' / provider (GB)	Downlink (B)
10^3	38.8	103.7	0.09	525,950
5×10^3	39.0	106.9	0.46	525,950
10^4	38.8	111.9	0.93	525,950
2×10^4	38.8	121.7	1.85	525,950
5×10^4	39.1	120.7	4.63	525,950
10^5	39.2	196.8	9.26	525,950
2×10^5	39.1	539.5	18.53	525,950
4.05×10^5	38.8	1,378.2	37.54	525,950

S5.6 Response packing, and what the trust split buys

Table S6 puts the two-ciphertext layout, the packed layout of Section 4.4, its compressed form and four-template responses through the reference implementation at both corpora's tag domains, together with baseline (vi), a single server holding the same encrypted columns. All rows come from one implementation on one machine and one thread, so they compare with each other and not with the C++ figures. Byte counts are a property of the parameters and of OpenFHE's serialisation, not of the machine; the C++ prototype's 262,975 B per ciphertext against 263,011 B here is a serialisation-version difference.

Table S6: Fast path per provider: two-ciphertext against packed layouts, and a single server on the same columns (baseline vi). Reference implementation, one thread; medians of 21 runs. k is the number of templates answered.

Layout	k	Provider (ms)	Downlink (B)
$N = 11{,}580$ *tags*			
Two ct (unpacked)	1	53.0	526,022
Packed, 1 ct	1	23.5	263,011
Packed + compressed	1	27.0	132,081
4× two ct	4	194.9	2,104,088
Packed, 1 ct	4	35.4	263,011
Packed + compressed	4	32.7	132,081
Single server (vi)	1	42.6	263,211
$N = 35{,}162$ *tags*			
Two ct (unpacked)	1	55.8	526,022
Packed, 1 ct	1	31.4	263,011
Packed + compressed	1	36.4	132,081
4× two ct	4	245.6	2,104,088
Packed, 1 ct	4	74.2	263,011
Packed + compressed	4	92.3	132,081
Single server (vi)	1	92.5	263,211

Packing halves the provider's work and the downlink. One rotate-and-sum over half-rows replaces two over full rows: a single template falls from 53.0 to 23.5 ms at $N = 11{,}580$ and from 55.8 to 31.4 ms at $N = 35{,}162$, and the response from two ciphertexts to one, 263,011 B, or 132,081 B compressed. Four templates in one compressed response cost 32.7 ms and 132,081 B at $N = 11{,}580$, against 194.9 ms and 2,104,088 B for four two-ciphertext responses (16× less downlink). At $N = 35{,}162$ they need 35 chunks instead of 12, so the cost grows to 74–92 ms while the response stays one ciphertext.

Against a single server, the trust split buys the client side. Baseline (vi) needs no non-collusion assumption. Its server costs 42.6 and 92.5 ms at the two tag domains, 1.8 and 2.9 times one packed CR-SHE provider (23.5 and 31.4 ms), because its contraction multiplies ciphertext by ciphertext; its response is the same one ciphertext. The ratios compare one provider with one server: two CR-SHE providers together spend 47.0 and 62.8 ms, so in total server work the trust split costs more than a single server at the smaller tag domain and saves about a third at the larger. Its query is not (Table S7): 789,033 and 2,367,099 B of fresh ciphertexts against a 260 or 294 B DPF key per provider (520 and 588 B in total for two), 3,035 and 8,051 times more uplink per provider, and 14.6 and 67.5 ms of client encryption against 0.54 and 0.62 ms of key generation. Query compression by ciphertext expansion [8] would cut that uplink to one ciphertext for more server automorphisms; we did not implement it.

On retrieval a single server must bring every word of the packed $N \times \lceil N_d/32 \rceil$ index into contact with an encrypted one-hot query, about $\lceil N \lceil N_d/32 \rceil / \ell \rceil$ plaintext-ciphertext products in a direct packed-selection formulation; query compression, preprocessing or specialised PIR layouts may change that. At the measured 0.092 ms per product this estimate is 81.6 ms at $N_d = 2 \times 10^4$ and 1.65 s at $N_d = 405{,}184$, against 26.0 ms and 1.10 s for CR-SHE's packed

Table S7: Per-query uplink and client cost: CR-SHE's two-party key against the single server's encrypted query. Reference implementation, one thread.

	Key (B)	Enc. query (B)	Cts	Encrypt (ms)
$N = 11{,}580$	260	789,033	3	14.6
$N = 35{,}162$	294	2,367,099	9	67.5

retrieval (Table S9); across the sweep the ratio is 1.5–3.1× for that cost model, not against a measured optimised single-server system, and the two figures come from separate runs. The separation on every single-keyword path is therefore a small constant factor in server work, set by a homomorphic slot operation (11 ns here) against a word operation, plus three orders of magnitude in uplink. That is the honest case for the non-collusion assumption on these paths: it pays where the client is thin or the link is metered, which is the IoT setting. On ranges (Section 7.2) the gap is wider.

S5.7 The flexible path

Table S8 reports the two rounds of a flexible query at $N = 11{,}580$ in the reference implementation, swept in decreasing N_d to control for warm-up. Round one, the retrieval, grows linearly with N_d as its $\Theta(NN_d)$ contraction predicts. Round two costs 20.1–35.6 ms with no trend in N_d over this range, because one to five packed products are dwarfed by the twelve rotations of the row sum, and it returns one 132,081 B ciphertext whatever the selection. Its uplink is a 16 B seed to $n-1$ providers and $8N_d$ B to one. Every flexible query in the reference implementation's tests, including conjunctions, disjunctions and ad-hoc weights, decrypts to the plaintext ground truth.

Table S8: Flexible path, $N = 11{,}580$, $n = 2$, reference implementation, one thread. Round-two uplink is the heavy provider's; the others receive 16 B.

N_d	Round 1 (ms)	Round 2 (ms)	Up (B)	Down (B)	I' (GB)
1,000	41.2	20.1	8,000	132,081	0.09
5,000	247.3	35.6	40,000	132,081	0.46
10,000	399.4	21.9	80,000	132,081	0.93
20,000	758.0	26.4	160,000	132,081	1.85

S5.8 Packed masked index

Table S9 compares the masked index with one membership bit per $\mathbb{Z}_p$ word against 32 bits per word, at the primary corpus's tag domain $N = 11{,}580$, in the reference implementation. Packing divides the index by exactly 32 and makes retrieval 19 to 27 times faster at $N_d = 10^4$ and 2×10^4, where both layouts fit in this machine's memory. At the full 405,184 records the packed index is 1.17 GB, against 37.5 GB unpacked (computed), and a retrieval takes 1.10 s and returns 101,296 B per provider. Every retrieval returned exactly the posting list.

S5.9 Multi-cloud deployment

With $n \in \{2, 3, 4\}$ separate provider processes and a client over TCP on one host (Table S10), provider compute is flat in n, at 34.82, 34.85 and 35.48 ms, so the client waits on the slowest provider, not on their sum. End-to-end latency rises from 44.1 to 52.5 ms, all of it transport of the n responses; the client's combine, verify and decrypt cost 1.37–1.63 ms. The transport

Table S9: Masked index, unpacked against 32 bits per word, $N = 11{,}580$. Reference implementation, one thread; retrieval per provider.

N_d	Layout	Index (GB)	Retrieval (ms)	Downlink (B)
10,000	unpacked	0.926	424.4	80,000
10,000	32 bits/word	0.029	22.0	2,504
20,000	unpacked	1.853	696.2	160,000
20,000	32 bits/word	0.058	26.0	5,000
100,000	32 bits/word	0.289	234.6	25,000
405,184	32 bits/word	1.173	1095.8	101,296

covers loopback only: wide-area latency, bandwidth variation, stragglers and recovery were not measured, so no fixed cross-region increment can be inferred from these figures.

Table S10: Single-host deployment over loopback, medians of 100 queries. Transport excludes wide-area latency.

n	End-to-end (ms)	Provider (ms)	Transport (ms)	Client (ms)
2	44.12	34.82	8.30	1.37
3	45.40	34.85	9.09	1.52
4	52.53	35.48	15.48	1.63

S5.10 Communication

At $N = 11{,}580$ the two-party construction sends a 266 B key to each provider, 532 B of uplink in total. The non-succinct n-party instantiation used for $n > 2$ sends 32 B to each of $n - 1$ providers and 92,656 B to one, so total uplink is 92,688 B at $n = 2$, 92,720 B at $n = 3$ and 92,752 B at $n = 4$. Tolerating more collusion is therefore nearly free in this instantiation, but moving beyond two providers at all costs a factor of 174 in uplink over the succinct two-party tree. A succinct $(n-1)$-private instantiation would reduce the heavy key from $O(N \log p)$ to $O(2^{n/2}\sqrt{N}\lambda)$.

Downlink is 525,950 B per provider for an aggregate in the two-ciphertext layout and 132,081 B packed and compressed, for up to four templates in our configuration, and a fixed $\lceil N_d/32 \rceil$ words for a retrieval, which is 25,000 B at $N_d = 10^5$ packed (800,000 B unpacked), none depending on $|S|$. A range query and a flexible round each return one 132,081 B ciphertext.

S5.11 Setup, masking and ingest costs

Unpacked, the mask costs a factor of 64 in index storage over a plaintext bitmap; packed, a factor of 2. In the C++ prototype the unpacked masked index at $N_d = 10^5$ took 5.6 s to build, the fast-path column A_f 1.2–12 ms per template, occupying 2 ciphertexts (0.53 MB) whatever N_d, and a complete provider bundle for the full corpus (context, evaluation keys, pk and eight columns) was 12 MB and took 0.8 s to produce. Insertion into the hot buffer ran at 4.5–14.1 million records per second; a fast-path static-tier rebuild cost 14.7–14.9 ms per template, 58.0 μs per insertion at $B = 256$ and 0.91 μs at $B = 16{,}384$, while the buffer scan added 98.8–100.0% to a query. The range cube registered in Section 7.2 took 1.3 s (combined) and 9.8 s (per-corner) to build and encrypt in the reference implementation.

S5.12 Core scaling and throughput

In the C++ prototype a single fast-path query does not speed up with cores (34.3, 34.4 and 38.4 ms at 1, 8 and 64 threads), because the homomorphic contraction dominates, but throughput does: 29.1 queries/s with one client, 56.9 with two, 110.2 with four, and a plateau of 121.6–127.7 from sixteen to sixty-four clients, with median latency rising from 34.4 to 516 ms; between four and sixteen clients throughput is within 15% of its maximum and median latency stays under 135 ms.

S5.13 Role-scoped access

Table S11 repeats the extraction attack of our earlier measurement, in which a client sending raw additive shares of basis vectors read every row of the masked index undetected (the C++ prototype recovered 512 of 512 rows). The reference deployment has 512 tags over three roles of 4,096 records each; the attacker holds role 0 and sends a raw share vector for every row. With one mask key it recovers every row exactly, including all 365,227 membership bits that belong to the other two roles. With role-scoped keys it recovers none of those bits and no row exactly; when it applies its own key to the other roles' words, 6.06% of them pass the width check, against the 6.25% expected of uniform words. Through the fast path the same user obtained, for all 512 tags, exactly the role-0 aggregate and in no case the all-role one, and a flexible round asking for every record returned the role-0 total. The price is storage: the aggregate columns, cube tables and record column are held once per role, while the masked index, the largest structure, is shared.

Table S11: Index extraction by a role-0 user sending raw shares for all 512 rows; three roles of 4,096 records. Reference implementation.

Index	Rows exact	Other roles' bits recovered	Other words pass width check
One mask key	512	365,227 of 365,227	n/a
Role-scoped keys	0	0 of 365,227	6.06%

S5.14 Second corpus: twelve sites and a larger tag domain

The primary corpus has three devices, so its tag universe is dominated by time buckets. The C++ prototype therefore repeated every experiment above on the Beijing corpus, whose twelve sites give three times the tag domain ($N = 35{,}162$, posting lengths median 12, mean 131.5, maximum 102,344). Missing readings get their own `:na` tag rather than falling into the lowest band, since CO is missing in 4.92% of records, and temperature is shifted by +50 °C so that it stays non-negative. Every correctness check passed unchanged, including a byte-identical 525,950 B downlink for $|S| = 0$ and $|S| = 5{,}000$.

The fast path is unaffected by the larger tag domain. It cost 40.2–41.2 ms from $N_d = 5 \times 10^3$ to 4.05×10^5 (Table S12), 5% more than on the primary corpus for 3.04× the tags: the packed products grow from 2 ciphertexts to 5, but the fixed rotate-and-sum, encryption and serialisation dominate. The first point, 54.6 ms at $N_d = 10^3$, carries the warm-up artefact of Section S5.5, since this sweep ran in increasing N_d. *The unpacked masked index is storage-bound.* The general path cost 382.0 ms at $N_d = 10^5$ against 196.8 ms on the primary corpus, and its unpacked index grew to 28.13 GB; the harness refused the last two points, which needed 56.26 and 113.98 GB, against a 47.5 GB budget. Packing (Section 4.3) divides all of these by 32, to 3.6 GB at $N_d = 4.05 \times 10^5$ (computed).

The remaining measurements track the primary corpus. Provider compute in the single-host deployment was flat in n (38.67, 37.65 and 38.47 ms for $n = 2, 3, 4$), with end-to-end latency

Table S12: Second corpus (N = 35,162, N_d up to 420,370): aggregate latency and unpacked index size, template SUM, C++ prototype, two-ciphertext layout. Sweep in increasing N_d; "n/a" marks a point the harness refused on its memory budget.

N_d	Fast (ms)	General (ms)	I' (GB)	Downlink (B)
10^3	54.6	106.3	0.28	525,950
5×10^3	40.4	86.9	1.41	525,950
10^4	40.2	99.7	2.81	525,950
2×10^4	40.5	145.5	5.63	525,950
5×10^4	40.7	220.8	14.06	525,950
10^5	41.2	382.0	28.13	525,950
2×10^5	40.9	n/a	56.26	525,950
4.05×10^5	40.9	n/a	113.98	525,950

rising from 48.17 to 56.04 ms in transport alone. The DORY-style search-only baseline cost 305.6 ms against 41.2 ms for the complete fast path, so the gap widens to 7.4× as N grows; the homomorphic equality scan, 24.4 s, is essentially unchanged, 592× the fast path. Throughput plateaued at 117–124 queries/s; insertion into the hot buffer ran at 4.5–11.1 million records per second; a fast-path static-tier rebuild cost 28.7–29.1 ms, twice the primary corpus and linear in N; and building the unpacked masked index at $N_d = 10^5$ took 16.6 s against 5.6 s, while A_f stays 5 ciphertexts (1.3 MB). Selection latency and keys depend on N alone and were not re-measured: the two-party key is 300 B and the n-party heavy key $8N + 16 = 281{,}312$ B, both computed.

S6. Withdrawn leakage figures from the C++ prototype

The C++ prototype's earlier figures are withdrawn. An earlier run of the attack in the C++ prototype reported higher unmasked recovery: 4.27% overall on the primary corpus, 100% in every non-temporal class of both corpora, 3.80% of one-minute windows and 0.31% of hourly windows. Several of these figures are not attainable by any attacker who sees only row statistics. Rows with the same observed statistic are indistinguishable to such an attacker, so whatever its strategy and auxiliary knowledge it expects at most one correct assignment per distinct value: a class whose n tags take v distinct lengths can lose at most v/n of them, and the whole index at most the number of distinct lengths over N. An hourly window of the second corpus holds at most one record per site, so its length is one of at most twelve values, whatever the corpus builder; in our build it takes seven, and at most 0.020% of the hourly windows can be re-identified, against the 0.31% reported. The twelve sites share four lengths, so at most 33.3% of them, not 100%. On the primary corpus the one-minute windows take twenty lengths (between 14 and 43 messages), a bound of 0.17% against 3.80%, and the whole index fifty-six, a bound of 0.484% against 4.27%. The reference attack meets these bounds exactly at $f = 1$. The C++ attack must therefore have used information beyond the lengths, most plausibly an order of tied candidates that followed the true labels. The bounds are computed from the per-tag lengths of our corpus build (`bench/leakage_bounds.py`, which also reproduces every unmasked, padded and volume-hiding rate from those lengths and fails when a single tie is planted); only the twelve-value bound on hourly windows holds for any build. We report the reference figures, which the released code and data reproduce. The conclusion is unchanged and is now measured against the standard defences: an unmasked replicated index exposes the meaningful tag classes, power-of-two padding still exposes many of them, and the masked index, like volume-hiding padding but without its storage cost, leaves the attack at random guessing.

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